MILITARY SPECIFICATION

INSECTICIDE, 2-PERCENT DIAZINON DUST

1. SCOPE

1.1 This specification covers one grade of insecticide, 2-percent diazinon dust, for use as a dusting powder for the control of roaches.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY
MIL-P-116 - Preservation, Methods of.

STANDARD

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Federal Insecticide, Fungicide, and Rodenticide Act.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N. Y.)

3. REQUIREMENTS

3.1 Composition. - The finished insecticide dust shall be an intimate mixture of either

(a) Technical diazinon (see 3.1.1) and diluent (see 6.3); or

(b) Diazinon 25W (see 3.1.2) and diluent (see 6.3).

The mixture shall contain not less than 2 percent by weight of actual diazinon (O,O-diethyl O-(2-isopropyl-4-methyl-6-pyrimidinyl)phosphorothioate.).

3.1.1 Technical diazinon. - The technical diazinon shall consist of not less than 90 percent of actual diazinon, determined as specified in 4.5.1.1, and not more than 15 percent of normal impurities incident to manufacture. Impurities shall have further specific limits as follow:

	Maximum percent by weight
Water (see 4.5.1.3)	0.05
Acidity (as H_2SO_4) (see 4.5.1.4)	0.03

3.1.2 Diazinon 25W. - Diazinon 25W shall contain not less than 25 percent of actual diazinon when tested as specified in 4.5.1.2. The remainder shall be normal impurities incident to the manufacture of diazinon, and the following diluents and formulating agents: calcium silicate, calcitic carbonate, magnesium carbonate, and wetting and dispersing agents.

3.2 Particle size of finished powder.

3.2.1 Average diameter. - The surface-mean particle diameter of the insecticide powder, by air-permeation test as specified in 4.5.2.1, shall be not greater than 5.0 microns.

3.2.2 Dry sieve test. - The finished insecticide shall be a free-flowing powder devoid of lumps. It shall be of such fineness that not less than 99 percent will pass a 177-micron (U. S. Standard No. 80) sieve in the dry sieve test as specified in 4.5.2.2.

3.3 Color. - The insecticide shall be of a light color, such as cream or white.

3.4 Workmanship. - The product shall be clean, uniform, and free from any defects which might impair its utility.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be

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kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. - For purposes of sampling and inspection a lot shall consist of all insecticide manufactured as one batch.

4.3 Examination of filled containers. - A random sample of filled containers shall be selected from each lot in accordance with Standard MIL-STD-105 at inspection level I and acceptable quality level - 2.5 percent defective to verify compliance with this specification regarding fill, closure, marking and other requirements not involving test. Examination shall be conducted as follows:

Examine	Defects
Net contents	Not as specified.
Container	Not type specified. Any leakage. Any break or dent which may affect serviceability of the container. Any unsound seam.
Marking:	
Label data and destination marking	Missing, illegible, incorrect or incomplete. Does not contain data specified in 5.1.3.1. Not lithographed, stenciled, or printed on main panel of the body.

4.4 Tests and examination of insecticide. - Insecticide shall be tested as shown in table I. The inspection level shall be I-3 in accordance with Standard MIL-STD-105. The lot size shall be expressed in units of filled containers. If any specimen fails in any test, the lot represented shall be rejected.

Table I - Tests

Characteristic	Requirement paragraph	Test paragraph	Sample size
Diazinon content	3.1	4.5.1.1, 4.5.1.2	1 pound
Water (when applicable)	3.1	4.5.1.3	1 pound
Acidity (when applicable)	3.1	4.5.1.4	1 pound
Particle size:		4	
Average diameter	3.2.1	4.5.2.1	1 pound
Dry sieve test	3.2.2	4.5.2.2	1 pound

4.4.1 Examination. The insecticide shall be examined for cleanliness, uniformity, color, free-flowing properties, and any evidence of foreign matter, to determine conformance with 3.3 and 3.4.

4.5 Test procedures. -

4.5.1 Chemical analysis. -

4.5.1.1 Diazinon content in technical diazinon. -

4.5.1.1.1 Preparation of 0.1 N perchloric acid in glacial acetic acid. - Combine 9 milliliters of perchloric acid (70 to 72 percent reagent grade) with 100 milliliters of glacial acetic acid in a 1-liter volumetric flask. Mix well, add 20 milliliters of acetic anhydride and make to the mark with additional glacial acetic acid. Allow to stand at least 24 hours. To standardize this solution, dissolve 0.2 gram of pure sodium carbonate in 50 milliliters of glacial acetic acid and titrate this solution with the solution of perchloric acid, using -naphtholbenzein as indicator. 0.200 gram of sodium carbonate corresponds to 37.73 milliliters of 0.1 N perchloric acid. Care shall be taken that all titration with perchloric acid be done in absolutely anhydrous media.

4.5.1.1.2 Procedure (technical diazinon). - Weigh accurately about 1 gram of the sample and transfer with 100 milliliters of ether to a 250-milliliter separatory funnel. Extract with three portions of 50 milliliters of distilled water, and then with successive portions of 0.05 N sodium hydroxide until the phenolphthalein indicator added to the water layer remains red. Finally, extract twice with 50-milliliter portions of distilled water. All the water washings are extracted separately and successively with two portion of ether each, using the same ether for all water washings. Combine these ether extracts with the first ether containing the main amount of the diazinon, dry with sodium sulfate, and filter. Distill off the ether, dissolve the residue in 20 milliliters of glacial acetic acid, and titrate with standardized 0.1 N perchloric acid in glacial acetic acid, using 1 percent -naphtholbenzein solution in benzene as indicator, until the solution becomes dark green.

where a = milliliters of 0.1 N perchloric acid
w = weight of sample in grams

Percent O,O-diethyl O-(2-isopropyl-4-methl-6-pyrimidinly) phosphorothioate

$$\frac{a \times 304.35 \times 100}{w \times 10000}$$

4.5.1.2 Diazinon content in insecticide dust or in diazinon 25W. - Weigh accurately an amount of the finished dust containing about 1 gram of technical diazinon and disperse it in ether. Decant through a fritted glass crucible (about 30-milliliter capacity) and repeat the dispersing six times to extract the diazinon quantitatively. Transfer the ether solutions

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to a 500-milliliter separatory funnel and proceed as specified in 4.5.1.1.2 starting with the water extractions described in the second sentence.

4.5.1.3 Water content in technical diazinon. -

Caution: Precaution shall be taken to prevent the apparatus, the reagents, and the material from being exposed to atmospheric moisture.

4.5.1.3.1 Apparatus. -

At least three 50-milliliter glass-stoppered Erlenmeyer flasks, which have been thoroughly cleaned, rinsed with water, and dried in an air oven at 110°C. (230°F.), will be required. The apparatus required for the determination of moisture content shall consist of a side-arm buret connected to an aspirator bottle which shall be used as the reservoir for the Fischer reagent (see 4.5.1.3.2). The nipple of the bottle shall be connected to the side arm of the buret. Air coming into each reservoir is first passed through concentrated sulfuric acid, and then through a U-tube containing a suitable dehydrating agent. Beyond the U-tube the air line is split with a T-tube; one branch of the T-tube is attached to the top of the buret and the other to the top of the reservoir. By this device, as the buret is filled with the Fischer reagent, air contained in it passes into the reservoir. As the buret is emptied, air from the atmosphere is drawn in through the drying-train. A one-holed rubber stopper of suitable size to fit the Erlenmeyer flasks is fitted over the buret tip so that the end of the tip extends through the rubber stopper. Titrations are carried out in 50-milliliter glass-stoppered Erlenmeyer flasks, held against this stopper. This prevents air turbulence in the neck of the flask and minimizes exposure of the solutions to air during titration. When the buret is not in use, the tip of the buret is always fitted with a dry Erlenmeyer flask to minimize evaporation and contamination of the solution in the buret.

4.5.1.3.2 Reagents. -

(a) Anhydrous methanol, containing not more than 0.1 percent by weight of water.
(b) Anhydrous pyridine, containing not more than 0.1 percent by weight of water.
(c) Sulfur dioxide, commercial grade, liquid, from a cylinder, or produced from sodium sulfite by adding sulfuric acid.
(d) Iodine, reagent grade, resublimed.
(e) Fischer reagent. - Dissolve 63.0 gram of iodine in 100 milliliters of dehydrated pyridine, cool in ice, and pass sulfur dioxide into the solution until a gain in weight of 32.3 grams has occurred, taking care to avoid absorption of atmospheric moisture. Add sufficient dehydrated methanol to produce 500 milliliters, allow to stand for 24 hours, and transfer to the reservoir of the buret.

(1) Standardization of Fischer reagent. - Introduce about 20 milliliters of dehydrated methanol into the Erlenmeyer flask and titrate with the Fischer reagent until the orange-yellow color changes to a distinct brown. (The volume of the Fischer reagent added need not be recorded. The brown color should persist for at least one minute with the flask held firmly against the rubber stopper and gently swirled.) Add to the contents of the Erlenmeyer flask 4 drops (120-140 milligrams) of distilled water, accurately weighed, from a 20-milliliter glass-stoppered dropping-bottle, which restores the orange-yellow color. Again titrate the solution in the flask with the Fischer reagent to the same end-point as before. Calculate the water equivalence (factor) of the Fischer reagent, as milligrams of water per milliliter of the reagent, from the weight of the water added.

NOTE: As the Fischer reagent deteriorates continuously, it should be standardized immediately before use or daily, as required. When freshly prepared, 1 milliliter of the reagent is equivalent to about 5 milligrams of water.

4.5.1.3.3 Procedure. - Weigh accurately about 2 grams of the sample into a perfectly dry Erlenmeyer flask fitted with an air-tight glass stopper. Introduce 20 milliliters dehydrated methanol into the flask. Connect the flask to a reflux condenser fitted with a calcium chloride guard-tube and heat the flask on a waterbath until the material dissolves in the methanol. Cool the flask to room temperature without removing the assembly. When the flask is cooled to room temperature, separate the flask from the assembly and titrate the solution contained in it with Fischer reagent until the orange-yellow color changes to brown. Carry out a blank determination on 20 milliliters of dehydrated methanol under the same conditions as given above.

4.5.1.3.4 Calculation. -

$$\text{Moisture content, percent by weight} \quad \frac{(V - v)f}{10\,w}$$

where V - volume of the Fischer reagent required for the titration of the solution (milliliters).
v volume of the Fischer reagent required for the blank determination (milliliters)

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f = factor of Fischer reagent (milligrams per milliliter)
w = weight of sample (grams)

4.5.1.4 Acidity of technical diazinon. - Dissolve 10.0 grams of the sample in 25 milliliters of acetone. Solution may be effected by gently warming if necessary. Add 75 milliliters of distilled water and titrate immediately with 0.02 N N_2OH ("a" milliliters), using methyl red as indicator. Carry out a blank determination on 25 milliliters of acetone and 75 milliliters of distilled water with 0.02 N N_2OH ("b" milliliters).

Percent acidity as H_2SO_4 = 0.0098 (a-b)

NOTE: The blank may take the form of a small titre with 0.02 N HCl ("c" milliliters), in which case the percent acidity as H_2SO_4 - 0.0098 (a-c).

Alternatively, the end-point may be determined electrometrically.

4.5.2 Particle size.-

4.5.2.1 Average diameter of insecticide dust. - The surface-mean particle diameter shall be determined by the air-permeation method. The apparatus used shall be a Government-approved air-permeation apparatus (see 6.2). The solid density of the powder, knowledge of which is prerequisite to particle fineness measurement by the air-permeation procedure, may be determined by any standard method. A suitable method is the conventional method of liquid immersion in a specific gravity bottle having a ground-in thermometer, with distilled water as the immersion liquid, with a wetting agent if needed. In the use of this method, care shall be taken to guard against significant loss of water-soluble material; the pycnometer (specific gravity bottle) shall have a nominal volume of 25 milliliters at least, and the weight of sample used for observation shall be as great as can be handled conveniently in the pycnometer used.

4.5.2.2 Dry-sieve test of insecticide dust. - Screen a dry 20-gram sample through a U. S. Standard sieve of the designated mesh size using a single-eccentric type mechanical shaker which imparts to the sieve a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Continue the screening for 15-minutes. Weigh the final residue remaining on the sieve and calculate the percentage passing through the sieve.

4.6 Inspection of preparation for delivery. - Sample packages and packs shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance to the requirements of Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use. -

5.1.1 Packaging. - Packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early use. The containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice and as follows:

5.1.3.1 Label data. - The following label data shall be durably and legibly marked on the main panel of the body of each unit container by lithographing, stenciling, or printing with a noncorrosive, weather-resistant, water-proof ink which will not smear nor rub off during handling:

"Stock No - - - -
Insecticide, Diazinon 2 Percent, Dust.
Net weight

ACTIVE INGREDIENT	
Diazinon*	2 percent
Inert ingredients	98 percent

*O, O-diethyl O-(2-isopropyl-4-methyl-6-pyrimidinyl) phosphorothioate

CAUTION

Only trained personnel should apply this dust. Toxic when taken internally, inhaled, or spilled on skin. Food, dishes, cooking utensils, and table tops must be protected from dust by a suitable covering. Clean up all visible deposits of dust which might be contacted by children or pets.

DIRECTIONS FOR USE

Apply dust to the interior of barracks and other buildings, paying particular attention to areas behind and beneath cabinets, refrigerators and sinks, and to cracks, crevices, and other areas where cockroaches are likely to crawl or rest. Use a plunger, bulb, or bellows duster to blow the dust into the areas to be treated. Do not apply dust to areas where it would be unsightly or would be removed by routine cleanup operations. One application, unless removed, will remain effective several weeks or more. Reapply as needed.

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MANUFACTURER'S NAME AND ADDRESS
DATE OF PACK
LOT NUMBER
CONTRACT OR ORDER NUMBER
DESTINATION MARKING"

5.1.3.2 The label shall be cleared for compliance with the Federal Insecticide, Fungicide, and Rondenticide Act. Application to the Pesticides Regulation Branch, U. S. Department of Agriculture, Washington 25, D. C., shall be made for registration of the label prior to shipment of the finished insecticide in interstate commerce. Additional markings shall be as specified in the contract or order.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.1).

5.2.1.1 Packaging. - Insecticide shall be packaged in 5-pound cans, 5-gallon (25-pound capacity) pails or 25-pound capacity fiber drums as specified (see 6.1).

5.2.1.1.1 Level A. -

5.2.1.1.1.1 Five-pound can. - The 5-pound can shall conform to Type V., class 2, round of Specification PPP-C-96. Exterior Plan B coating and side seam striping shall be required. Cans shall be provided with wire handles which shall be galvanized or protectively coated to resist corrosion.

5.2.1.1.1.2 Five-gallon pail. - The 5-gallon pail shall conform to Type II (lug cover), class 3 of Specification PPP-P-704.

5.2.1.1.1.3 Fiber drum. - The fiber drum shall conform to Type II, grade A of Specification PPP-D-723. Drums, unless otherwise specified in the contract or order, shall be provided with a barrier, lining or an interior coating which will not affect or be affected by the product packaged.

5.2.1.1.2 Level C. - Packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use.

5.2.1.2 Packing. -

5.2.1.2.1 Level A. -

5.2.1.2.1.1 Cans. Cans shall be arranged and packed for overseas shipment in accordance with the requirements of the appendix to Specification PPP-C-96. Cans packed in tiers shall have a W6 (Specification PPP-B-636) fiberboard pad placed between the tiers.

5.2.1.2.1.2 Pails and fiber drums. - Pails or fiber drums will require no additional packing. When specified (see 6.1) pails or fiber drums shall be palletized in accordance with Standard MIL-STD-147.

5.2.1.2.2 Level B. -

5.2.1.2.2.1 Cans. - Cans shall be arranged and packed for domestic shipment in accordance with the requirements of the appendix to Specification PPP-C-96. Cans packed in tiers shall have a fiberboard pad of 200-pounds hoisting strength, conforming to class 1 of Specification PPP-B-636 placed between the tiers.

5.2.1.2.2.2 Pails and fiber drums. - Pails or fiber drums will require no additional packing. When specified (see 6.1) pails or fiber drums shall be palletized in accordance with Standard MIL-STD-147.

5.2.1.3 Marking. - In addition to any special marking required by the contract or order or herein, shipping containers and palletized unit loads shall be marked in accordance with Standard MIL-STD-129.

5.2.1.3.1 Label data. - Label data shall be as specified in 5.1.3.1 and 5.1.3.2.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Packaging, packing or marking requirements other than those required by paragraph 5.1 (see 5.2).

6.2 Air-permeation apparatus. - An approved apparatus is the self-calculating air-permeation apparatus of the U. S. Department of Agriculture (Patent No. 2,261,802), or equal. The Sub-Sieve Sizer, manufactured by the Fisher Scientific Company, may be used as the test instrument. Each individual apparatus is subject to inspection by the procuring activity as a check on calibration and performance.

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6.3 Diluent. - The following diluents have been found to be suitable for formulating diazinon dusts:

Diluent	Availability
Emtal 23-A	Eastern Magnesium Talc Company Baldwin Avenue, South Burlington, Vt.
Canadian Talc No. 5 Virginia Talc No. 9 Georgia Talc No. 2387 Caldust A No. 1887	Whittaker, Clark & Daniels, Inc. 260 West Broadway, New York 13, N. Y.
Pyrax ABB	R. T. Vanderbilt Company, Inc. 230 Park Avenue, New York 17, N. Y.
Calwhite T Dilcarb 9W	Georgia Marble Company 838 Board Street, Bank Blvd. Trenton 8, N. J.
Mississippi Dust Diluent	Mississippi Lime Company St. Genevieve, Mo.
R. M. Whiting Soapstone 290 No. 320 Talc	Commercial Minerals Company, Inc. 310 Irwin Street San Francisco 7, Calif.
CCC Diluent	Calcium Carbonate Corporation Chicago, Ill.
Glendon Pyrophyllite	Glendon Pyrophyllite Company Greensboro, N. C.
Kalvent No. 2 Snowflake Dricalite	Thompson-Weinman Company 52 Vanderbilt Avenue New York, N. Y.
Mistron 139	Sierra Talc and Clay Company 1888 Huntington Drive South Pasadena, Calif.
J-R Cassadil	Alex. M. McIver & Son P. O. Box 156 Charleston, S. C.
Vertal No. 20	Vermont Talc Company Chester, Vermont
MIL Slip Talc	Milwhite Company Houston, Texas
Whiting 400	Kennedy Mineral, Inc. Los Angeles, Calif.
SER-Y	Summit Mining Corporation Aspers, Pa.
V-Dust	L. H. Butcher Company 19th and Vermont San Francisco, Calif.

Precautions: Attapulgite, Kaolin and other clays generally used in formulating pesticides cause diazinon to decompose rapidly and, therefore, should not be used in formulating diazinon dusts.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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